

First-year Analysis Examination
Part One
May 2018

Answer FOUR questions in detail.
State carefully any results used without proof.

1. Let $f : [0, 1] \rightarrow [0, 1]$ be a (weakly) increasing function. Prove that

$$\alpha := \inf\{t \in [0, 1] : f(t) \leq t\}$$

exists and is fixed by f . Show further that α is the *smallest* fixed point of f .

2. Let M be a metric space, let $U \subseteq M$ be open and let $A \subseteq M$ be arbitrary. Prove that $\overline{A} \cap U \subseteq \overline{A \cap U}$. Can this inclusion be strict? Explain.

3. Let $(a_n : n \geq 0)$ be a sequence of real numbers. Assume that there exists $\lambda \in (0, 1)$ such that $|a_{n+1} - a_n| \leq \lambda|a_n - a_{n-1}|$ for each positive integer n . Prove that the sequence $(a_n : n \geq 0)$ converges.

4. Let $f : X \rightarrow Y$ be a continuous bijection between metric spaces. For each of the following statements, decide whether true or false, giving proof or counterexample as appropriate:

- (i) if X is compact, then $f^{-1} : Y \rightarrow X$ is continuous;
- (ii) if Y is compact, then $f^{-1} : Y \rightarrow X$ is continuous.

5. Let the real-valued function f be differentiable on the open interval $(0, 1)$. Show that if its derivative f' is bounded, then f extends to a continuous function on the closed interval $[0, 1]$.
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